

The Market for Housing Regulation: what has been built, what it established, and what is open

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September 2026

Abstract

A chronicle, not a paper: one document to reconstruct the project’s state from, front to back. It lists every memo with what later work revised in it, walks through them in the order they were finished, inventories the data, states the model the project is building toward, maps each model object to the data that would identify it, and keeps the register of what is open.

13 memos and 304 pages rest on one dataset of 16,199 decisions. Every item the Planning Commission heard from 1998 to 2026 with a case number is extracted into structured fields (a model at 96.6% field accuracy on a frozen test half), dated to its hearing, and followed into DBI’s permit record, the Planning Department’s records, the Commission’s own motions and conditions, the parcel map, prices, fees and the law.

Several headline claims did not survive later work, and the timeline records each. The model case: “no amount of work recovers condition content before 2010” was a claim about two URL patterns; a census of every case found text for 2.4% and 7.6% of the conditioned items of the two pre-2010 eras — little, not none. Others: the two linkage routes do overlap; two of the project’s own join keys were wrong; most of the apparent slowness of Commission-linked permits is immortal time; the developer’s clock on the DR track starts at the DBI filing.

τ is now measured in dollars; σ is measured as dispersion, not yet as the model’s primitive. A new building of ten or more units owes a median \$61k per unit in fees and inclusionary obligation; the developer’s clock has a coefficient of variation of 0.79 on the conditional-use track and 0.95 off the Commission. What blocks a first draft is identification: separating τ from σ , the deterred projects, and treating σ as a parcel-level primitive (§8).

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1 The question

The market for housing regulation. A developer who wants to build in San Francisco buys, in effect, a permission from the city, and the price has two parts. One is deterministic and written down: fees, the inclusionary obligation, the template conditions every approval carries. The other is stochastic: how long the process takes and what the Commission will ask for. The project’s premise is that the second part matters for what gets built, and where.

The real-options frame. A parcel is an option to develop. Uncertainty over delay and conditions raises the value of waiting, so it raises the developer’s filing threshold; and because the cost of waiting is paid on the project’s whole value, the threshold rises by different amounts on different parcels. Variance in the regulatory process therefore does two things a deterministic cost of the same expected size does not: it deters filing, and it re-routes the filings that happen toward higher-value parcels.

How the evidence is assembled. The Commission’s minutes, scraped for 1998–2026, are parsed into agenda-item blocks; a deterministic stage dates each block to its hearing and derives the sitting’s attributes; a language model with retrieved examples extracts each item’s fields. From the item, four routes lead outward: to DBI’s building permits (a permit number printed in the minutes, the Planning records’ own permit field, or a parcel-and-date rule), to the Planning Department’s records (case number), to the parcel map, assessor roll, prices and zoning (block and lot), and to the Commission’s own motions, where the conditions of approval are written. The law — fees, the inclusionary program, state statutes — enters as quoted claims checked against their primary sources. Three standards run through all of it: extraction rules are frozen and hashed before they are validated, and in-sample and out-of-sample figures are reported apart; a machine–label disagreement is checked against the source before the machine is called wrong; and no number in a memo is typed by hand.

Two papers. Paper 1 estimates the within-jurisdiction mechanism in San Francisco, where the Commission’s record goes back to 1998 and can be followed item by item. Paper 2 takes the primitives estimated there to the fragmentation question across the Bay Area’s jurisdictions. Everything in this chronicle serves Paper 1.

2 The memos, in order

Table 1 lists every memo by the date it was first completed, the question it was asked, its answer in a sentence, and what later work revised or overturned in it. The dates are those first recorded each memo (for the three written in this batch, the run log’s); every memo has since been revised in place, and the revisions column is the point of the table.

Table 1: Every memo, in order of completion: the question it was asked, its answer in a sentence, and what later work revised or overturned in it.

Date	Memo	Question	Answer	Revised or overturned by
2026-09-04	Meeting-level information (§3.1)	Can the hearing date and the sitting’s attributes be derived from the minutes?	Dates are right for every gold item; the attribute fields score in the mid-90s out of sample, worst in 1998–2001.	2026-09-04: the 2018 scraping gap refilled. 2026-09-10: the round-4 figure (held-out meetings) and the all-meetings figure (93.9%, adjudicated) are different measurements; a name-reducer defect had split six commissioners into two each.

Date	Memo	Question	Answer	Revised or overturned by
2026-09-07	Item-level extraction (§3.2)	Rules or a language model for the item fields?	Haiku with retrieved examples, 96.6% against the regex’s 75.2% on the frozen test half.	2026-09-10: the regex baseline had been understated by its own scoring; few-shot retrieval had let an item answer itself, which touched every “all items” figure but not the test half.
2026-09-07	The corpus memo (§3.3)	What does the item-level dataset contain?	Denial has nearly vanished while conditioning tripled; delay’s upper tail lengthened; DR’s share of the docket collapsed.	2026-09-10: the conditioning rate was recomputed after a defect fix. 2026-09-11: the case universe fixed at 8,958; its distinct-parcel count is superseded by the acquisition memo’s join.
2026-09-08	Permit linkage (§3.4)	How far can a decision be followed into DBI’s permits?	The printed permit number matches on 97.6% of 2,295 numbers; the parcel fallback runs at 0.72 precision; the Commission’s conditions are not in DBI.	2026-09-11: the fallback’s precision was measured on discretionary reviews only, so the permit content memo keeps that tier (T3) apart; the Planning records’ permit field is a second route.
2026-09-08	Conditions of approval (§3.5)	Where do the conditions of approval live, and how much is obtainable?	In the Commission packet’s draft motion, addressable by case number from 2011; templated, and a thirteen-category scheme spans them.	2026-09-08 and 2026-09-11: “the two routes never cover the same item” withdrawn (the records’ permit field reaches 46.3% of conditional uses); “no amount of work recovers condition content before 2010” withdrawn — it was a claim about two URL patterns, and the census found text for 2.4% and 7.6% of the pre-2010 eras: little, not none.
2026-09-08	Predicting delay (§3.6)	How much of the hearing-to-hearing time is forecastable?	The level somewhat, the tail barely; the residual dispersion is the first proxy for the cost of uncertainty.	2026-09-11: the case universe note. Its clock starts at the first hearing; the acquisition memo’s filing clocks and the clocks memo’s full distributions supersede it as the duration measure.
2026-09-08	Data acquisition (§3.7)	Where are the denominator, the clock, the scale, the price and the fees?	A defined parcel-year risk set, a filing clock, a price on every parcel-year, and fees that vary across sections rather than years.	2026-09-08: two of its own conclusions reversed (the lettered-lot key; the two-digit year). 2026-09-11: on the DR track the developer’s clock starts at the DBI filing, not the Planning record (from the permit content memo).
2026-09-11	DataSF catalogue (§3.8)	What does DataSF hold under “planning” that the pipeline does not use?	No dataset of Commission actions exists; review-time metrics and DBI’s station routing are the unused delay data.	2026-09-11: the feasible envelope added.
2026-09-11	Conditions census (§3.9)	What do the conditions say, across every case and motion?	Condition text for 48.2% of conditioned items; a template with a thin project-specific layer; the entitlement’s term is a 36-month template.	None yet.
				2026-09-11 (this brief): the parser’s third round; the document once read as the earliest motion in the universe was two motions bound together, and every span resting on it moved; the numbers inside the conditions extracted.

Date	Memo	Question	Answer	Revised or overturned by
2026-09-11	Permit content (§3.10)	Are the projects that reach the Commission different from similar ones?	Bigger and slower, but most of the time difference is immortal time; the lower abandonment survives.	2026-09-11 (this brief): the landmark comparison quantified how little of the time gap survives; cost placeholders dropped; the first-construction-document date dropped as an outcome.
2026-09-11	Clocks (§3.11)	What are the full distributions of every duration, by track and parcel value?	Long and dispersed on every track; about a third never record completion; dispersion does not fall consistently with value.	Extended the same day with the outcome chain.
2026-09-11	Exactions (§3.12)	What inclusionary requirement and impact fees vested for each project?	A sourced version chain and vesting rules; a median \$61k per unit for new buildings of ten or more units.	None yet; its open questions are stated in it.
2026-09-11	Regulatory timeline (§3.13)	Which state laws and ordinances generate variation this data can measure?	6 of 45; the opt-in pathways are elections, not assignments.	None yet.

Every supersession, collected

Table 2 collects every place where later work replaced a claim or a figure, including the brief’s own premises that this batch checked. Most are corrections of the project’s own instruments — keys, parsers, scoring — found when a number was checked against its source; the rest are claims that generalised beyond what had been measured.

Table 2: Every supersession found: what a memo claimed or measured, and what replaced it. Old and new figures are read from the logs and the brief that record them.

Date	Memo	Was	Now
2026-09-04	Meeting level	2018 was a corpus hole of 2 documents; series crossed it	refilled from the S3 packet prefix to 42 documents
2026-09-07	Extraction	accuracy against the hand labels as the measure	52% of 187 judged disagreements were label errors; unadjudicated accuracy is a lower bound
2026-09-07	Extraction	gold records complete	a scalar/list mismatch had silently dropped lot numbers on 122 of 232 gold records; repaired
2026-09-08	Permit linkage	the permit route collapsed after 2019	a change in how the minutes print the number; reading it recovers 711 items and 483 permits
2026-09-08	Conditions of approval	discretionary review carries no conditions section	1 of 41 discretionary reviews do
2026-09-08	Conditions of approval	the two linkage routes never cover the same item	the Planning records’ permit field reaches 46.3% of conditional uses
2026-09-08	Data acquisition	parcel join below the acceptance criterion, at 89.5%	97.7% once lettered lots and multi-block items are keyed correctly
2026-09-08	Data acquisition	pre-2002 records do not go back that far under any key	expanding the two-digit year recovers 527 of 530 pre-2002 cases
2026-09-08	Data acquisition	a first price surface	it carried a +0.90 log bias, caught by held-out scoring and rebuilt
2026-09-10	Extraction	the regex baseline at 70.7%	75.2% once scored through the same storage layer as the model
2026-09-10	Meeting level	146 distinct roll-call names	78 once a mojibake token stopped splitting commissioners in two
2026-09-10	Permit linkage and the analyses	the parcel-key fix applied	the lettered-lot bug was still live in four scripts; the fix grows the fallback’s validation set from 2,909 to 3,274
2026-09-11	Conditions of approval	no condition content before 2010	2.4% and 7.6% of the pre-2010 eras’ conditioned items have text

Date	Memo	Was	Now
2026-09-11	The corpus memo	8,295 distinct parcels	8,980 from the acquisition memo’s join (not yet regenerated)
2026-09-11	Conditions	the earliest motion read for a case in the universe was heard in 1994	two motions bound together; the earliest such item now dates from 1999
2026-09-11	Permit content	the within-cell time gap between linked and unlinked permits, read at face value	most is immortal time: measured from the hearing, about 9% survives in means and 28% in medians
2026-09-11	Data acquisition	the DR clock starts at the Planning record	at the DBI filing, a median 150 days earlier
2026-09-11	Exactions (in its own build)	the 2019 register as cited	the archive served the 2020 file for the 2019 capture; the fetcher now refuses another capture
2026-09-11	The brief	about 1,040 pre-2010 items with a motion number and 590 without; 50,366 addresses	1,032 and 554; 127,920 addresses
2026-09-11	The brief	AB 1763 among the Housing Accountability Act amendments	its own title is a density-bonus bill
2026-09-11	The brief	a 25-unit ordinance of July 2026	none among the Board’s 2026 ordinances through Ord. 0133-26

3 A walk through them

Each subsection says what the memo asked, what it established, what it did not, and what it changed downstream. Detail is in the memos.

3.1 Meeting-level information (2026-09-04)

Asked. The item needs two facts it does not carry: the day it was heard, and the character of the sitting. Both belong to the meeting. *Established.* A deterministic date stage that never re-derives block boundaries and was right on every gold item in four frozen rounds; attribute fields in the mid-90s out of sample; a meeting table of 1,219 sittings over 1998–2026. Scored today against all 134 hand-confirmed meetings, the attributes agree on 93.9% of 938 values, 91.0% for 1998–2001 and 98.3% from 2015. *Not established.* The all-meetings figure is unadjudicated, and spot checks find hand labels that are wrong; two commissioners named Lee are left unresolved rather than guessed. *Downstream.* The date is the join key to every other panel, and the hearing sequence of a case — first hearing, continuances, final action — is built from it; the meeting table also carries who sat, which the corpus memo’s commissioner analysis uses. The method’s rule that a corpus hole is drawn as a gap rather than interpolated became a standing rule of the project.

3.2 Item-level extraction (2026-09-07)

Asked. Rules or a language model for the item fields? *Established.* On a frozen test half of a 294-item gold set, a small model with retrieved examples scores 96.6% against the regular expressions’ 75.2%; adjudicating the disagreements found that about half were label errors, so an accuracy measured against unadjudicated gold is a lower bound. *Not established.* The modern era is thin in the gold set. *Downstream.* The corpus pass that produced the item table, and the rule that segmentation stays deterministic while field semantics go to the model. Its storage layer (raw, interim and clean replies kept apart) is what later made a silent data loss recoverable. Two of its figures were corrected in the code review that followed (Table 2).

3.3 The corpus memo (2026-09-07)

Asked. What does the item-level dataset contain? *Established.* Across 16,199 items and 8,958 cases: outright denial has nearly vanished while conditioning has more than tripled — the Commission substituted a conditioning technology for a denial technology; repeat appearances are a large share of the docket and the upper tail of time from first to last hearing lengthened; discretionary review’s share of the docket collapsed after the 2013 reforms, as that memo dates them. *Not established.* Any ex-ante measure of burden: the data records realised outcomes. *Downstream.* It set the agenda: conditions (what is in them), linkage (what happens after), delay (how much is forecastable). Its closing section named four routes from realised outcomes to an ex-ante measure of burden; the real-options frame the later memos adopt is the one that took. Its distinct-parcel count is superseded and not yet regenerated.

3.4 Permit linkage (2026-09-08)

Asked. How far can a decision be followed into DBI’s permits? *Established.* The printed permit number matches on 97.6% of all 2,295 distinct numbers the minutes print; the apparent collapse of the route after 2019 was a change in how the minutes print the number, not in the data. The route reaches discretionary reviews and almost no conditional uses; a parcel-and-date fallback narrows the candidates at 0.72 precision and 0.92 recall. Across DBI’s rows, 87 mention a Commission condition at all. *Not established.* The fallback’s precision outside discretionary review, where it was measured. *Downstream.* The permit content memo keeps the fallback (T3) apart from the printed number (T1) and the Planning records’ permit field (T2), and calls only the first two “Commission-linked”. The negative result — conditions are not in DBI — sent the conditions work to the Commission’s own documents. The check that caught the post-2019 parsing failure, comparing the rate at which a thing is mentioned with the rate at which it is parsed, became a standard diagnostic.

3.5 Conditions of approval (2026-09-08)

Asked. The item table records that a project was conditioned, not how. Where is the content? *Established.* Not in the minutes, not in DataSF, not in DBI; in the Commission packet’s draft motion, whose Exhibit A numbers and names each condition, addressable by case number from 2011. The conditions are templated, and a thirteen-category scheme reaches most impositions; staff’s conditions mostly change a project’s obligations, the Commission’s amendments mostly change the building. *Overtured.* Two claims. “The two routes never cover the same item” was about what the minutes print; the Planning records’ permit field reaches 46.3% of conditional-use items. And “no amount of work recovers condition content before 2010” generalised from two URL patterns: the census found 2.4% and 7.6% of the pre-2010 eras’ conditioned items with text, against 78.9% for 2011–2014, and 1,586 of the 1,600 pre-2010 items without text are documents found at no address. *Downstream.* The census, which replaced its sample with every case; its taxonomy remains unvalidated against hand coding.

3.6 Predicting delay (2026-09-08)

Asked. How much of the time from a case’s first hearing to its last can be forecast from what a developer sees on the day of the first? *Established.* The level somewhat, the tail barely, and in-sample fit roughly halves out of sample; the staff recommendation is the most informative signal; the residual dispersion is large, and it is the first proxy for the cost of uncertainty. *Not established.*

Anything causal. *Superseded.* Its clock starts at the first hearing; the data-acquisition memo’s filing clocks and the clocks memo’s distributions replace it as the measure of duration.

3.7 Data acquisition (2026-09-08)

Asked. Where are the denominator, the clock, the scale, the price and the fees the real-options frame needs? *Established.* A defined risk set of 173,325 parcels and 4,261,894 parcel-years carrying 9,240 hearings (21.7 per ten thousand); a parcel join at 97.7% and a case join at 98.7%; two filing clocks that differ by a median of 0 days; a price on every parcel-year (out-of-sample RMSE 0.69 in logs where imputed); fees that vary across sections within a year (16.4% of the log-rate variance) rather than over years (3.0%). *Revised by itself.* Two conclusions reversed within days: the lettered-lot key and the two-digit year had been diagnosed as data problems and were our own keys. *Revised later.* On the DR track the developer’s clock starts at the DBI filing of the permit under review, a median 150 days before the Planning record (a dated note, 2026-09-11). *Extended.* The feasible envelope (§3.14).

3.8 DataSF catalogue (2026-09-11)

Asked. What does the portal hold under “planning” that the pipeline does not use? *Established.* 273 assets; no dataset of Commission actions, published or not; 82 assets flagged as possibly useful and not cached, of which the Planning Department’s record-level review-time metrics and DBI’s station routing are delay data nobody has measured against. *Downstream.* None used yet; the review-time metrics are in the next steps.

3.9 The conditions census (2026-09-11)

Asked. Replace the sample of twenty cases a year with every case and motion, and read what the conditions say. *Established.* 127,920 addresses probed, 10,165 documents found; condition text for 48.2% of the 4,048 conditioned items, the earliest heard in 1999. The conditions are a template with a thin project-specific layer, and what varies is numbers. The entitlement’s term is written into the template: a 36-month validity on 96% of the motions that state one. The parser was validated in frozen rounds; the third (this batch) scores 98.2% precision and 98.5% recall in sample and 96.7% out of sample on rows heard 2016–2025. *Revised in this batch.* The document once read as the earliest motion for a case in the universe was two motions bound together (Motion 13693, May 26, 1994, and a later one); every year-span resting on it moved. *Downstream.* The records request, the numbers table, and the exactions memo’s cross-check of the inclusionary schedule against the percentages the conditions print.

3.10 Permit content (2026-09-11)

Asked. What is in DBI’s table, and do the projects that reach the Commission differ from similar ones? *Established.* 5.0% of permits are development-relevant, and 7.8% of those are Commission-linked. Linked projects are bigger and slower, but a permit linked through a hearing held while it was pending had to survive to the hearing; measured from the hearing on permits alive at it, about 9% of the within-cell gap in time survives in means and 28% in medians. The lower abandonment of linked permits does survive (20.9% against 31.7% expired or withdrawn). *Revised in this batch.* The landmark comparison replaced a claim that almost none of the gap survives; cost placeholders were dropped; the first-construction-document date was dropped as an outcome for coverage.

3.11 Clocks (2026-09-11)

Asked. The full distribution of every duration, by track, parcel value, size, period and neighbourhood, with censoring handled. *Established.* Kaplan–Meier medians of the developer’s whole clock of 2,232 days (conditional use), 1,817 (DR), 3,394 (large projects) and 1,334 (ministerial); about a third of spells record no completion ten years on (36% on the conditional-use track); the coefficient of variation does not fall consistently with parcel value. The outcome chain (added the same day): of housing projects entitled 2005–2014, 79% of conditional uses and 63% of ministerial permits record a completion. *Not established.* σ as the model defines it: a cross-sectional CV mixes ex-ante uncertainty with what the developer knew.

3.12 Exactions (2026-09-11)

Asked. What inclusionary requirement and impact fees vested for each project? *Established.* A version chain from 2002, every rate and vesting rule a quotation re-checked against its source (394 claims); a median obligation of \$61k per unit for new buildings of ten or more units; bunching just below every inclusionary threshold (18 buildings of nine units against 5 of ten since Proposition C). *Not established.* The schedule agrees with the percentage the Commission’s conditions print in 53% of comparable cases; area-specific rates and the grandfathering deadline’s application are open.

3.13 The regulatory timeline (2026-09-11)

Asked. Which laws generate variation this data can measure? *Established.* 45 laws inventoried with quoted citations; 6 usable, 28 not measurable here; the opt-in pathways are elections, not assignments. The docket’s residential conditional uses fell from 29.4 a quarter to 22.7 after the Housing Production ordinances, so docket series need an era split there. *Not established.* Any effect.

3.14 The rest of this batch

The feasible envelope (a section of the data-acquisition memo). *Asked.* The risk set models non-filing, and “could have filed” needs a capacity; the counterfactual needs a benchmark that can be relaxed. *Established.* For every parcel-year: the height limit and bulk district read from the height district’s name, the lot area read at the map lot so that a condominium’s unit parcels share their building’s lot, the density rule of the zoning district in force, the rear yard, and a maximum unit count, with the state density bonus apart. The rules are quotations from the zoning control tables and the Government Code, checked like the exaction claims. Under central assumptions the read districts allow 795,840 units in 2026, and 687,630–937,959 across the sensitivity runs. *Not established.* Rules before and after the reading of the tables it rests on; the named neighbourhood commercial districts. *Downstream.* The risk set’s capacity and the counterfactual’s benchmark.

The numbers inside the conditions (a section of the conditions census). *Asked.* The census found that conditions are a template and that numbers are what vary; extract them. *Established.* 6,970 conditions with typed fields — validity periods, operating hours, bicycle and parking counts, stated inclusionary percentages and unit counts, notice radii and periods, monitoring intervals, screening dimensions — and their distributions over time. *Downstream.* The exactions memo’s cross-check of its schedule against the percentages the Commission’s conditions print.

The outcome chain (a section of the clocks memo). *Asked.* Conditional on entitlement, what share of approved projects and units is built, and how long does it take? *Established.* Completion shares by track and parcel value for projects entitled 2005–2014; cumulative incidence of completion with exit as a competing event. *Not established.* Units delivered are a lower bound: a completion recorded under another of a project’s permit numbers is missed.

The records request. *Asked.* Turn the census’s pre-2010 residual into a request. *Produced.* A draft letter to the Planning Department for the 1,032 conditioned items heard before 2010 whose motion number is known and the 554 whose is not, with a statement of everything already searched (127,920 addresses), and a separate set of questions for the Assessor-Recorder on recorded notices of special restrictions. Nothing has been sent.

4 What the data say so far

The substantive findings, collected from the memos that own them, each stated as the memo states it and each a description.

- **The Commission conditions rather than denies.** Outright denial has nearly vanished over the corpus while the share of items approved with conditions more than tripled (the corpus memo); 48.2% of the 4,048 conditioned items now have their condition text (the census).
- **The conditions are a template.** Most imposed conditions carry one of a few dozen headings; what varies is numbers. The entitlement’s term is part of the template: a 36-month validity on 96% of the motions that state one.
- **Delay is long, dispersed and hard to forecast.** From first filing to completion the Kaplan–Meier median is 2,232 days on the conditional-use track and 1,334 off the Commission, with coefficients of variation of 0.79 and 0.95 (the clocks memo); what a developer observes at the first hearing forecasts the level of hearing delay somewhat and its tail barely (the delay memo). About a third of spells record no completion ten years on.
- **Commission projects differ from similar ones mostly in what they survive.** Linked permits are bigger and slower, but most of the time difference is immortal time; their lower abandonment (20.9% against 31.7%) survives the correction (the permit content memo).
- **Most entitled housing is built.** Of housing projects entitled 2005–2014, 79% of conditional uses, 78% of discretionary reviews and 94% of large project authorisations record a completion; 17%, 16% and 3% exited (the outcome chain).
- **The written price is large and sits at thresholds.** A new building of ten or more units owes a median \$61k per unit, 20% of DBI’s valuation, mostly the inclusionary fee; new buildings are fewer just above each inclusionary threshold than just below it (the exactions memo).
- **The docket changed character after the Housing Production ordinances.** Residential conditional uses fell from 29.4 to 22.7 a quarter; SB 330’s five-hearing cap does not show in the hearing counts (1.9% of residential cases heard more than five times before, 2.4% after) (the regulatory timeline).

- **Fees vary across the city, not over time.** A year effect absorbs 3.0% of the variance in log fee rates; the variation is across sections and size bands (the acquisition memo).

5 The data inventory

Table 3 lists every table and panel the project has built or cached, its unit of observation and size, its years, its known defects and the memo that documents it. Everything lives under `$MFHR_DATA_ROOT`, not in git.

Table 3: The data inventory: every table and panel, its unit and size, its years, its known defects, and the memo that documents it.

Object	Unit (rows)	Years	Known defects	Memo
Minutes corpus and item table	item (16,199)	1998–2026	fields extracted by model at a measured error; modern era thin in the gold set	corpus; extraction
Meeting table	meeting (1,219)	1998–2026	attributes weakest in 1998–2001 (91.0% against 98.3% from 2015)	meeting level
Case universe	case (8,958)	1998–2026	case numbers as printed, two-digit years expanded	corpus; acquisition
Permit matches	printed permit number (2,295)	1998–2026	the printed route reaches DR, almost never CU	permit linkage
DBI permits, every column	permit (1,149,352)	1901–2026	cost placeholders;	permit content
Linkage tiers T1/T2/T3	item–permit link	1998–2026	first-construction date sparse T3 precision measured on DR only	permit content
Planning records	record	1920–2026 (sparse early)	record open date is not acceptance	acquisition
Parcel-year zoning panel	parcel-year (5,824,609)	1998–2026	zoning after 2016 carried forward	acquisition
Risk set	parcel-year (4,261,894)	1998–2026	condominium rule drops 1,301 hearings	acquisition
Price panel	parcel-year	1998–2026	imputed where no sale, flagged	acquisition
Fee registers	rate by year	2011–2026 (no 2017)	pre-2011 missing	acquisition; exactions
Conditions census and parse	condition row (94,169)	1984–2026 (thin early)	pre-2010 documents largely not online	conditions census
Conditions numbers	condition (6,970)	as the census	numeric fields only where stated	conditions census
Clocks panel	case or permit (34,320)	1985–2026	right-censoring; non-completion truncation	clocks
Outcome chain	housing project (10,253)	1985–2026	units delivered are a lower bound	clocks
Exaction panel	new-construction project (3,574)	2001–2026	priced from 2011; floor area mostly assumed; tenure mostly unknown	exactions
Law inventory and series	law (45)	enacted 2016–2025	18 effective dates inferred	regulatory timeline
Feasible envelope	parcel-year (5,824,609)	1998–2026	rules held at one reading; some districts unread	acquisition
Records request package	item (1,032 + 554)	1998–2009	draft; not sent	records request

6 The model as currently specified

The project is building toward a minimal model, deliberately narrower than the earlier `dr_supply_model` blueprint (a document this repository does not hold, so its contents are not summarised here):

One developer, one parcel, an option to file. The regulatory process imposes a deterministic cost τ (fees, inclusionary obligations, the template conditions) and a stochastic delay and outcome with variance σ . The exercise threshold depends on both, and the comparative statics differ across parcel value: τ shifts the threshold roughly uniformly in project value, σ shifts it differentially. The counterfactual holds expected burden fixed and sets σ to zero.

What each piece of data does. In this model the data play four parts. The exaction panel and the conditions census measure τ : the dollars and the template obligations a project owes, fixed at dates the law states. The clocks and the outcome chain measure the distribution of delay and of outcomes that σ summarises. The risk set, the price panel and the envelope give the parcels that could have filed, what they were worth, and how much they could hold — the population on which the filing threshold is estimated. And the regulatory timeline and the exaction panel’s version chain supply the changes in the rules that might move τ or σ separately. The comparative statics the model delivers — τ roughly uniform in project value, σ differential — are what make the parcel-value gradient of filing the central cross-sectional prediction, which is why every duration and obligation is reported by parcel value.

Deferred. To Paper 2 and to later drafts: strategic neighbours, endogenous supply of discretionary review, the Commission’s best response, and endogenous zoning. None of these is in the model above, and none of the data built so far is organised around them.

An assumption, not a finding. σ is currently treated as a parcel-level primitive estimated from a hazard. That is an assumption to be relaxed. The clocks memo measures cross-sectional dispersion within cells; whether that dispersion is uncertainty the developer faced, or heterogeneity the developer knew about, is not yet separated.

7 Which data object identifies which model object

Table 4 maps each primitive to the moment that would identify it, the dataset that supplies the moment, and whether that dataset exists. The two rows where the answer is not yet “yes” are the ones the open register calls blocking.

Table 4: Which data object identifies which model object.

Model primitive	Identifying moment	Dataset	Exists?
τ : fees and inclusionary obligation	dollar obligation per project, by vesting date	exaction panel	yes (priced from 2011)
τ : the template conditions	count and kind of conditions per motion	conditions census	yes (thin before 2010)
σ : delay dispersion	CV and Kaplan–Meier quantiles of each clock, by track and value	clocks panel	yes (cross-sectional, not ex ante)
σ : outcome risk	probability of denial, conditioning, abandonment	item table; outcome chain	yes
Arrival of discretion	share of projects that face a hearing, by parcel	risk set + linkage tiers	partly (T3 unvalidated outside DR)
Filing threshold	filing hazard by parcel value	risk set, price panel, planning records	yes
Parcel value	price per parcel-year	price panel	yes (imputed where no sale)
Capacity (the risk set’s “could have filed”)	maximum units per parcel-year	feasible envelope	yes (assumptions stated)
Option term	validity period and extensions	conditions numbers	yes
Separating τ from σ	variation in τ with σ fixed, and the reverse	inclusionary changes; SB 330; Housing Production ordinances	described, not estimated
Deterred projects	non-filing on the risk set	risk set	defined; selection not solved

8 The open register

Every unresolved item, with what it needs — data, a records request, a modelling decision or a literature check — and whether it blocks a first presentable draft.

The three that block. *Deterred projects*: the risk set models non-filing on parcels with capacity (the envelope now supplies it), but the parcels whose owners would have filed absent the regulatory process are not observed, and the filing hazard is estimated on the parcels that did. *Separating τ from σ* : the exactions memo and the regulatory timeline name variation in τ with σ plausibly fixed (the inclusionary changes and their filing-date cutoffs) and variation in σ (SB 330’s hearing cap, the Housing Production ordinances); none is yet an estimate. *σ as a primitive*: as §6 says.

Table 5: The open register: what is unresolved, who or what resolves it, and whether it blocks a first presentable draft.

Open item	Needs	Blocks draft
Deterred projects: the risk set models non-filing, but who would have filed is unobserved	modelling decision	yes
The unconstrained-envelope counterfactual: an envelope exists; its relaxation is unspecified	modelling decision	no
Identification of τ separately from σ : the candidate variation is described, not estimated	modelling decision	yes
σ is a parcel-level primitive estimated from a hazard — an assumption, not a finding	modelling decision	yes
The pre-2010 motion residual	records request	no
Recorded Notices of Special Restrictions: document type, bulk access, pre-1990 terms	records request	no
Whether T3 linkage can be validated outside discretionary review	data	no
Area-specific inclusionary rates, TSF grandfathering, existing-use credits, the 2017 register	data	no
How the 2016 grandfathering deadline was applied (conditions contradict the codified text)	data	no
Effective dates of 18 statutes inferred, not read	literature check	no
The Priority Equity Geographies polygon, for the Housing Production ordinances’ treated set	data	no
The meeting-level gold set is unadjudicated for its all-meetings figure	data	no
The 25-unit ordinance of July 2026 named in the brief	literature check	no
Review-time metrics and DBI station routing (DataSF), unused	data	no

9 Next steps

In priority order, with a rough sense of scale.

1. **Write down the estimator for the filing hazard** on the risk set, with τ from the exaction panel, the envelope as capacity, and parcel value from the price panel; decide how σ enters (a cell-level dispersion, or a latent parcel-level primitive). *Weeks.*
2. **Turn the inclusionary cutoffs into a first design:** the notch and the filing-date cutoffs, with the envelope held fixed. *Days to a first cut; weeks to an estimate.*
3. **Join the Priority Equity Geographies polygon** and cut the Housing Production ordinances’ treated set. *Days.*
4. **Validate T3 linkage outside discretionary review** on a hand-checked sample. *Days.*
5. **Send the records request** (after editing) and the Assessor-Recorder questions. *Days to send; months to answer.*
6. **Close the exaction panel’s gaps:** area-specific rates, TSF grandfathering, existing-use credits, the 2017 register. *Days each.*
7. **Use DataSF’s review-time metrics and DBI’s station routing** for the delay inside the permit. *Weeks.*

10 Established and not established

Established.

1. An item-level record of the Commission’s decisions, 1998–2026, dated and followed into permits, records, motions, parcels, prices, fees and law.
2. A dollar measure of τ from 2011 and a distributional measure of the durations on every track.
3. Which earlier claims were revised, and how (Table 1).

Not established.

1. σ as the model defines it.
2. Any separation of τ from σ , and any correction for deterred projects.
3. Condition content before 2010 beyond the thin share now read.

11 Reproducing this

```
cd code/commission_minutes_processing
python build_chronicle.py      # macros and tables for this memo
```

It reads the other memos’ generated macros files and tables, so each memo’s own pipeline runs first (each memo’s “Reproducing this” section gives its commands); the dates come from git and `run_log.md`, the page counts from the compiled PDFs and the row counts from the data.